

RESTRAINED LENGTHS FOR HDPE/PVC TRANSITIONS

The following tables for SDR 11 HDPE installations suggest restrained length requirements for the PVC side of an HDPE to PVC transition at a bridge, road, or stream crossing. This would be on each "side".

The tables reflect conservative temperature, PE modulus, and soil parameters and applied a method in common use to calculate the design loads.

Delta T = 45 Degrees "F".....this is the difference in the maximum and minimum temperatures to be experienced.

HDPE Modulus @ 40 Degrees "F" = 170,000 psi

Soil Type = GM (Silty gravels, gravel-sand-silt mixtures)

To use a different SDR multiply the table data by the ratio of the table SDR to the new SDR.

To use a different Delta T and a different PE Modulus multiply the table data by the ratio of the new parameter to the table parameter.

To use a different soil type consult table for adjustment multiplier and apply to table of restrained lengths.

Restrained Length of PVC (ft) to Restrain an SDR 11 HDPE Road/Bridge/Stream Crossing			
Pipe Diameter	Depth of Cover		
	24"	30"	36"
3"	32	26	22
4"	38	31	26
6"	54	43	37
8"	70	57	48
10"	83	68	57
12"	98	80	68
14"	111	91	77
16"	124	103	87

Coarse Grained Soils		Fine Grained Soils	
GW	Well-graded gravels and gravel-sand mixtures, little or no fines. Adjustment Multiplier = .9x	CL	Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, lean clays (backfilled using native soil). Adjustment Multiplier =.6x
SW	Well graded sands and gravelly sands, little or no fines. Adjustment Multiplier = .9x	ML	Inorganic silts, very fine sands, rock flour, silty or clayey fine sands (backfilled using native soil). Adjustment Multiplier = 1.4x
GP	Poorly graded gravels and gravel-sand mixtures, little or no fines. Adjustment Multiplier = 1.0x		
SP	Poorly graded sands and gravelly sands, little or no fines. Adjustment Multiplier = 1.0x		
GM	Silty gravels, gravel-sand-silt mixtures. Adjustment Multiplier = 1.0x		
SM	Silty sands,sand silt mixtures. Adjustment Multiplier =.7x		
GC	Clayey gravels, gravel-sand-clay mixtures. Adjustment Multiplier =.7x		
SC	Clayey sands, sand-clay mixtures. Adjustment Multiplier = .7x		